

FUNCTIONS OF BILE

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- Most of the functions of bile are due to **bile salts**.

1. DIGESTIVE FUNCTION

- Bile salts are required for the **digestion and absorption of fats** in the small intestine.

Emulsification of Fats

- **Emulsification** : Fat globules are broken down into minute droplets in the form of a milky fluid called **emulsion** in the small intestine.
- **Mechanism** :—
 - Lipolytic enzymes of the GI tract cannot digest large fat globules because fats are insoluble in water due to high surface tension.
 - Bile salts emulsify fats by **reducing surface tension** due to their natural detergent action.
 - Emulsification by bile salts **requires the presence of lecithin**.
- **Clinical Significance** : Unemulsified fat passes through the intestine and is **eliminated in feces**.

2. ABSORPTIVE FUNCTIONS

- **Absorption of Fats :—**

Bile salts combine with digested fats



Form water-soluble complexes called **Micelles**



Fats in the form of micelles are easily absorbed across the intestinal mucosa

3. EXCRETORY FUNCTIONS

- Bile serves as an important excretory route for:
 - a. **Bile pigments** (Bilirubin and Biliverdin)
 - b. **Heavy metals** (Cu^{2+} and Fe)
 - c. **Bacteria** (e.g., *Typhoid bacilli*)
 - d. **Toxins**
 - e. **Cholesterol**
 - f. **Lecithin**
 - g. **Alkaline phosphatase (ALP)**

4. LAXATIVE ACTION

- **Laxative** : An agent that induces defecation.
- Bile salts act as mild laxatives by **stimulating peristaltic movements** of the intestine.

5. ANTISEPTIC ACTION

- Bile inhibits the growth of certain harmful bacteria in the lumen of the intestine by its **natural detergent action**.

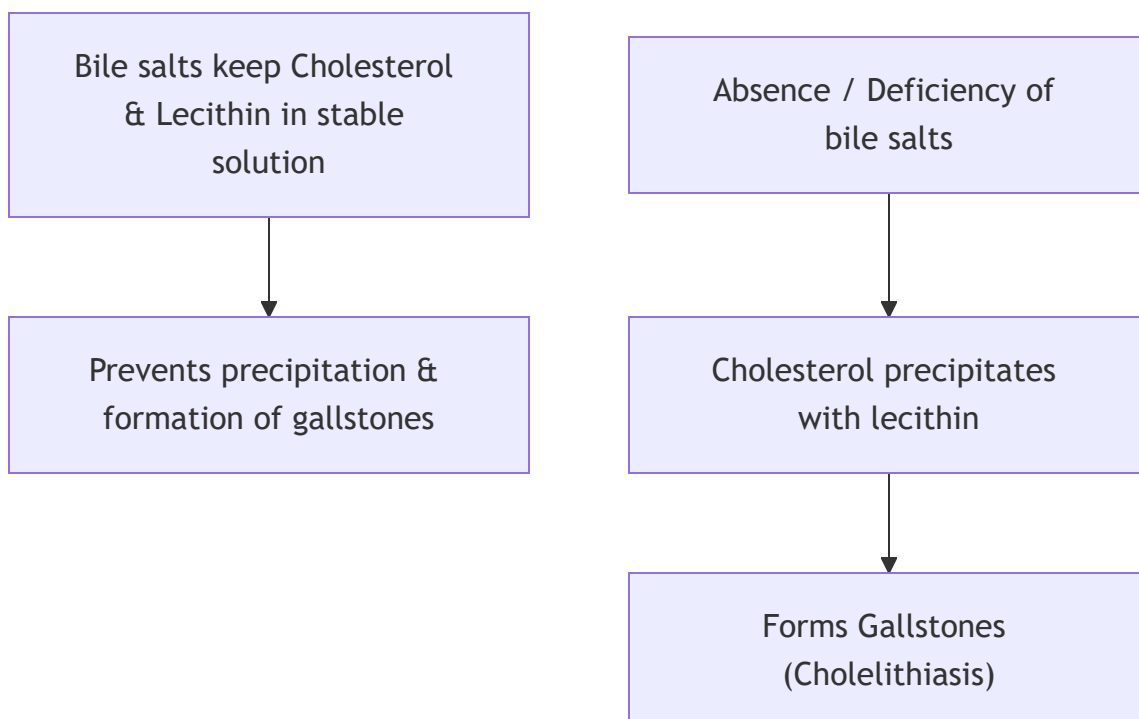
6. CHOLERETIC ACTION

- Bile salts stimulate the liver cells (hepatocytes) to **increase the secretion of bile**.

7. MAINTENANCE OF pH IN GI TRACT

- Alkaline bile helps **neutralize acidic chyme** entering the duodenum from the stomach, maintaining an optimal alkaline pH for pancreatic and intestinal enzymes.

8. PREVENTION OF GALLSTONE FORMATION



9. LUBRICATION FUNCTION

- **Mucin present in bile** acts as a protective lubricant for chyme as it moves along the intestinal tract.

10. CHOLAGOGUE ACTION

- **Cholagogue** : An agent that causes contraction of the gallbladder and release of stored bile into the duodenum.
- **Mechanism :—**

Bile salts act as cholagogues



Stimulate secretion of **Cholecystokinin (CCK)** from duodenal mucosa



Causes strong contraction of gallbladder and relaxation of Sphincter of Oddi